



LAB 03-EXPLORING EXCHANGE 2019 CONSOLES



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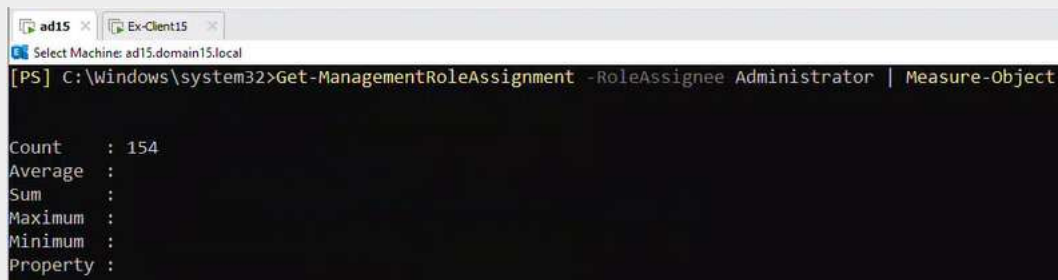
Exercise 1 – Exploring Exchange Administration Center (EAC)

Objective: Explore the Exchange Admin Center (EAC) console.

Tasks:

1. Remove the **NAT network card** from the server VM settings then **restart the server**.
2. Log in to the **adXX** server using **domainXX\administrator** account.
3. Open a web browser and access the **Exchange Administration Center (EAC)**.
4. Answer the following questions based on your observations in the EAC:
 - a. How many mailboxes are currently configured on your Exchange server?
There is one mailbox currently configured on the Exchange server.
 - b. What is the email address associated with the Administrator account?
Administrator@domain15.local
 - c. Is the Administrator a member of any distribution groups or email groups?
The Administrator account is not a member of any distribution groups or email groups.
 - d. Does the Administrator account have any associated contacts?
No, the Administrator account does not have any associated contacts.
 - e. How many role assignments does the Administrator have?

154



```

PS C:\Windows\system32>Get-ManagementRoleAssignment -RoleAssignee Administrator | Measure-Object

Count      : 154
Average    :
Sum        :
Maximum    :
Minimum    :
Property   :
  
```

- Which role grants permission to export or copy mailbox databases?
Mailbox Import Export
 - Which role authorizes configuration of compliance and auditing features?
Compliance Management
- f. What are the contents of the retention policy currently applied on your Exchange server?
Default MRM Policy
- g. Does your Exchange organization share information with other Exchange organizations (e.g.,

via federation)?

No, the Exchange organization does not currently share information with other Exchange organizations through federation.

- h. What is the antimalware policy behavior when malware is detected in an incoming or outgoing message?

The default antimalware policy is enabled and has the lowest priority. When malware is detected in an incoming or outgoing message, Exchange deletes the entire message. No sender notifications or administrator notifications are configured.

- i. Are there any mobile devices currently registered and synchronized via ActiveSync?

No, there are no mobile devices currently registered or synchronized via ActiveSync.

- j. Are there any existing public folders configured in your organization?

No, there are no existing public folders configured in the organization.

- k. What is the name of the active mailbox database used by your server?

Mailbox Database 0499151451

- l. How many digital certificates are installed on the server? 3

- For each certificate, indicate which Exchange services (e.g., SMTP, IIS, POP, IMAP) it is bound to.

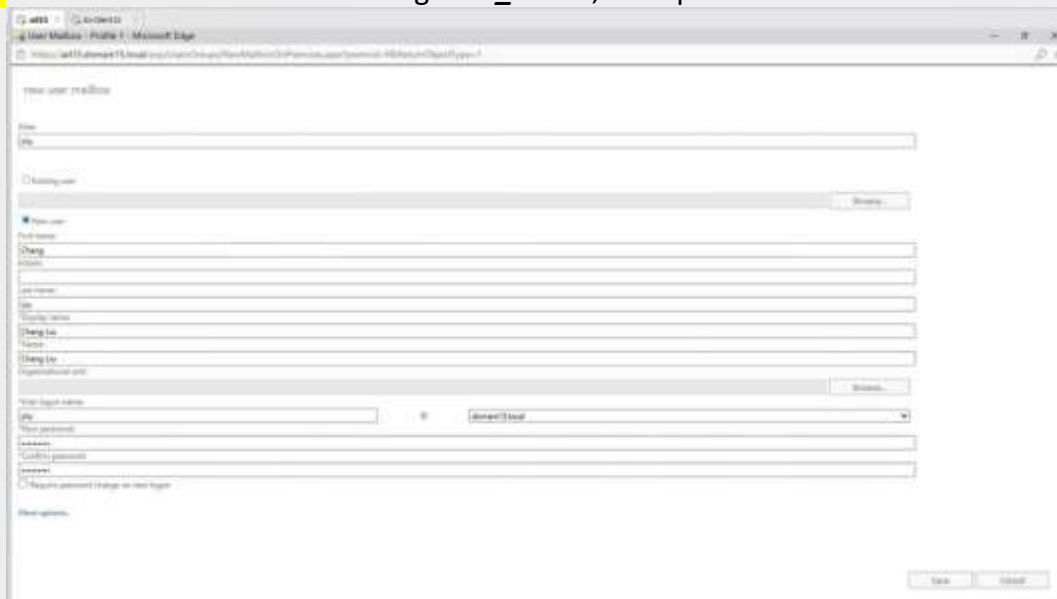
Certificate 1: Microsoft Exchange Server Auth Certificate Bound services: SMTP

Certificate 2: Microsoft Exchange Bound services: IMAP,POP,IIS,SMTP

Certificate 3: WMSvc-SHA2 Bound services: None

5. Using the "recipients" menu, create a new user with a new mailbox.

6. Create a new user mailbox using **Your_Name**, example:



7. Click **Save**.



DISPLAY NAME	MAILBOX TYPE	EMAIL ADDRESS
Administrator	User	Administrator@domain15.local
Chang Liu	User	cliu@domain15.local

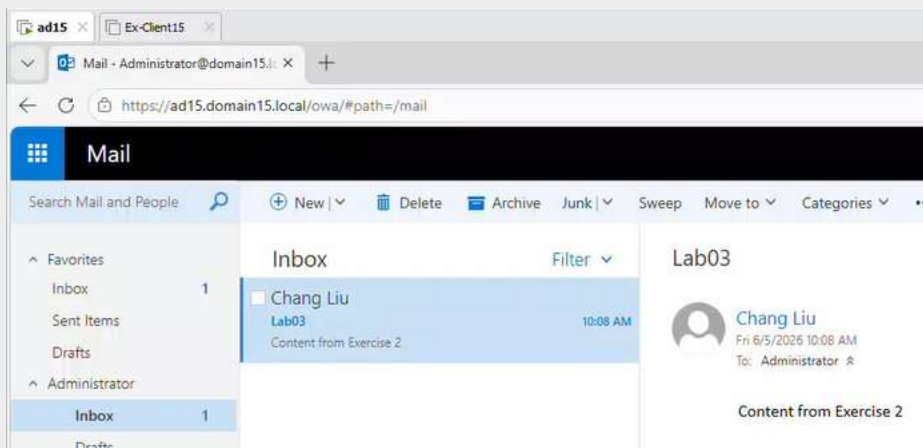
Exercise 2 – Exploring Outlook on the Web (OWA)

Objective:

- Explore the **Outlook Console on the Web (OWA)**

Tasks:

1. Open the **Outlook Web App (OWA)** using the new user account created during the previous exercise.
2. Compose a new email:
 - Recipient: **Administrator**
 - Subject: **Lab03**
 - Message Body: **Content from Exercise 2**
3. Send the email.
4. **Sign out** from OWA.
5. Log back in using the **Administrator** account.
6. Check that the Administrator account has successfully received the email sent by the new user.



7. Using the **administrator** account, open the **Exchange Administration Center (EAC)**.
8. Click on **recipients** then **Your_Name** then click on the **pencil** in the menu to edit.
9. In the **EAC**, click on **Recipients**.
10. Locate and select your newly created user (**Your_Name**).
11. Click the **pencil icon** to edit the **mailbox settings**.
12. In the mailbox **properties**, click on **Mailbox Usage**.
13. Review the **current mailbox quota**, which should be set to **2 GB** by default.
14. Click on More options modify the **quota** with the following settings:

ad15 Ex-Client15

Edit User Mailbox - Profile 1 - Microsoft Edge

https://ad15.domain15.local/ecp/UsersGroups/EditMailbox.aspx?pwmcid=4&ReturnObjectType=1&id=a373746b

Chang Liu

general

mailbox usage

contact information

organization

email address

mailbox features

member of

MailTip

mailbox delegation

Last login:

6/5/2026 10:07 AM

Mailbox Usage shows how much of the total mailbox quota has been used. Use this page if you wa

693.65 KB used, 0% of 2 GB.

Use the default quota settings from the mailbox database

Customize the quota settings for this mailbox

*Issue a warning at (GB):

1

*Prohibit send at (GB):

1

*Prohibit send and receive at (GB):

1.2

Use the default retention settings from the mailbox database

Customize the retention settings for this mailbox

*Keep deleted items for (days):

30

Don't permanently delete items until the database is backed up

15. Click **Save** to apply the new mailbox quota settings.

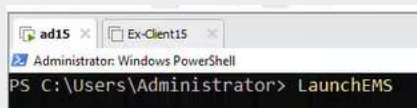
Exercise 3 – Exploring Exchange Management Shell (EMS)

Objective:

- Explore the **Exchange Management Shell (EMS)** console.

Tasks:

- Launch **Windows PowerShell**
- Load the Exchange Management Environment.



- Display all Exchange-specific cmdlets available.

ad15 Ex-Client15

Microsoft Windows [Version 10.0.17134.0] (c) 2016 Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> Get-ExchangeCommand

CommandType	Name	Version	Source
Function	Add-ADPermission	1.0	ad15.domain15.local
Function	Add-AvailabilityAddressSpace	1.0	ad15.domain15.local
Function	Add-ContentFilterPhrase	1.0	ad15.domain15.local
Function	Add-DatabaseAvailabilityGroupServer	1.0	ad15.domain15.local
Function	Add-DistributionGroupMember	1.0	ad15.domain15.local
Function	Add-FederatedDomain	1.0	ad15.domain15.local
Function	Add-GlobalMonitoringOverride	1.0	ad15.domain15.local
Function	Add-IPAllowlistEntry	1.0	ad15.domain15.local
Function	Add-IPAllowlistProvider	1.0	ad15.domain15.local
Function	Add-IPBlocklistEntry	1.0	ad15.domain15.local
Function	Add-IPBlocklistProvider	1.0	ad15.domain15.local
Function	Add-MailboxDatabaseCopy	1.0	ad15.domain15.local
Function	Add-MailboxFolderPermission	1.0	ad15.domain15.local
Function	Add-MailboxLocation	1.0	ad15.domain15.local
Function	Add-MailboxPermission	1.0	ad15.domain15.local
Function	Add-ManagementRoleEntry	1.0	ad15.domain15.local
Function	Add-PublicFolderClientPermission	1.0	ad15.domain15.local
Function	Add-ResubmitRequest	1.0	ad15.domain15.local
Function	Add-RoleGroupMember	1.0	ad15.domain15.local
Function	Add-ServerMonitoringOverride	1.0	ad15.domain15.local
Function	Clear-ActiveSyncDevice	1.0	ad15.domain15.local
Function	Clear-MobileDevice	1.0	ad15.domain15.local
Function	Clear-TextMessagingAccount	1.0	ad15.domain15.local
Function	Compare-TextMessagingVerificationCode	1.0	ad15.domain15.local
Function	Complete-MigrationBatch	1.0	ad15.domain15.local
Function	Connect-Mailbox	1.0	ad15.domain15.local

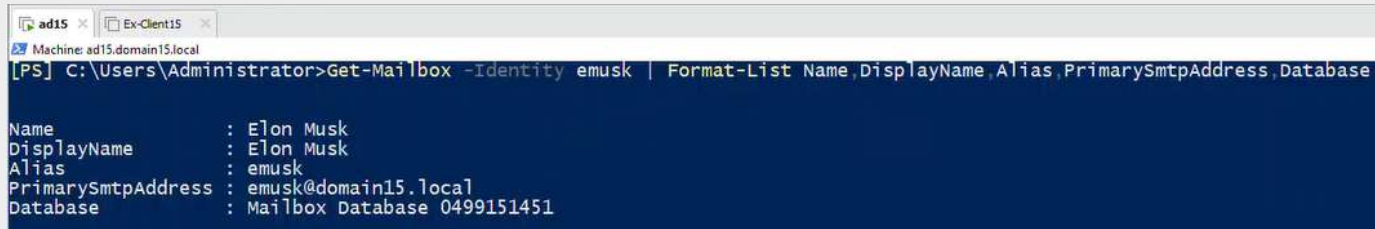
4. Using the **New-Mailbox** cmdlet, create a new mailbox for the user "Elon Musk".



```
[PS] C:\Users\Administrator>New-Mailbox -Name "Elon Musk" -UserPrincipalName emusk@domain15.local -Password (ConvertTo-SecureString -String 'Password123' -AsPlainText -Force) -FirstName Elon -LastName Musk -Alias emusk
```

Name	Alias	ServerName	ProhibitsSendQuota
Elon Musk	emusk	ad15	Unlimited

5. Verify that the mailbox has been created.



```
[PS] C:\Users\Administrator>Get-Mailbox -Identity emusk | Format-List Name, DisplayName, Alias, PrimarySmtpAddress, Database
```

```
Name           : Elon Musk
DisplayName     : Elon Musk
Alias          : emusk
PrimarySmtpAddress : emusk@domain15.local
Database       : Mailbox Database 0499151451
```

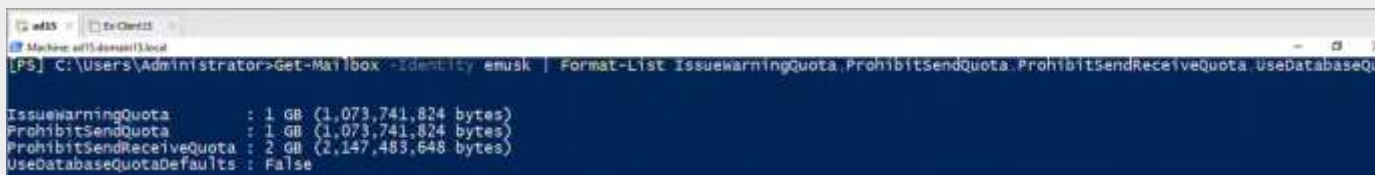
6. Using the **Set-Mailbox** cmdlet, configure the mailbox quotas for Elon Musk as follows:

- Issue a warning when reaching **1 GB**.
- Prohibit sending when reaching **1 GB**.
- Prohibit sending and receiving when reaching **2 GB**.
- Ensure database quota defaults are disabled
(Use the parameter: **-UseDatabaseQuotaDefaults \$false** at the end of the command.)



```
[PS] C:\Users\Administrator>Set-Mailbox -Identity "Elon Musk" -IssueWarningQuota 1gb -ProhibitSendQuota 1gb -ProhibitSendReceiveQuota 2gb -UseDatabaseQuotaDefaults $false
```

7. Verify that the quota settings have been correctly modified.

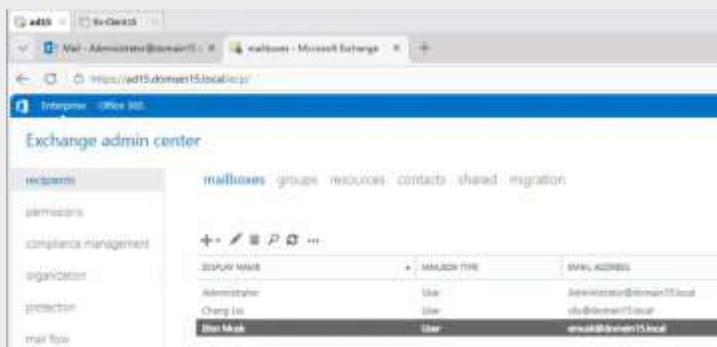


```
[PS] C:\Users\Administrator>Get-Mailbox -Identity emusk | Format-List IssueWarningQuota, ProhibitSendQuota, ProhibitSendReceiveQuota, UseDatabaseQuotaDefaults
```

```
IssueWarningQuota      : 1 GB (1,073,741,824 bytes)
ProhibitSendQuota      : 1 GB (1,073,741,824 bytes)
ProhibitSendReceiveQuota : 2 GB (2,147,483,648 bytes)
UseDatabaseQuotaDefaults : False
```

8. Log in as the **Administrator** to the **Exchange Administration Center (EAC)**.

9. Verify in EAC that the new user and mailbox have been created.



10. Confirm that the configured quotas match the settings applied in **Step 6**.

ad15 Ex-Client15

Edit User Mailbox - Profile 1 - Microsoft Edge

https://ad15.domain15.local/ecp/UsersGroups/EditMailbox.aspx?pwmcid=8&ReturnObjectType=1&id=0723074d-fbf8-44a1

Elon Musk

general

mailbox usage

contact information

organization

email address

mailbox features

member of

MailTip

mailbox delegation

Last logon:

Mailbox Usage shows how much of the total mailbox quota has been used. Use this page if you want to change

0 B used, 0% of 1 GB.

☐ Use the default quota settings from the mailbox database

☒ Customize the quota settings for this mailbox

*Issue a warning at (GB):

1

*Prohibit send at (GB):

1

*Prohibit send and receive at (GB):

2

☒ Use the default retention settings from the mailbox database

☐ Customize the retention settings for this mailbox

*Keep deleted items for (days):

14

☐ Don't permanently delete items until the database is backed up

Exercise 4 – Exploring Permissions in the Exchange Administration Center (EAC)

Objective:

- Understand the structure and purpose of **Admin Roles** and **User Roles** in Exchange Server 2019, and how to view and interpret role assignments.

Tasks:

- Log in to the **EAC** using an Administrator account.
- In the left-hand menu, click on **Permissions**.
- Notice that you have two main sections:
 - Admin Roles
 - User Roles**
- Under **Admin Roles**, review the list of existing **Role Groups**.
- Select 3 Role Groups and answer the following:
 - What is the name of the role group?
Organization Management, Recipient Management, Compliance Management.
 - What roles are assigned to this group?
Organization Management: Members of this management role group have permissions to manage Exchange objects and their properties in the Exchange organization. Members can also delegate role groups and management roles in the organization. This role group shouldn't be deleted.

Recipient Management: Members of this management role group have rights to create, manage, and remove Exchange recipient objects in the Exchange organization.

Compliance Management: This role group will allow a specified user, responsible for compliance, to properly configure and manage compliance settings within Exchange in accordance with their policy.

- Which users or groups are members of this role group?

Organization Management: Administrator

Recipient Management: Administrator

Compliance Management: Administrator

Example to analyze: **Organization Management, Recipient Management, Compliance Management.**

6. Select the **Organization Management** role group.

7. Click on the **pencil icon** () to edit it.

8. Explore the following:

- What roles are assigned to this group?

Active Directory Permissions

Address Lists

Audit Logs

Cmdlet Extension Agents

Compliance Admin

Data Loss Prevention

Database Availability Groups

Database Copies

Databases

Disaster Recovery

Distribution Groups

Edge Subscriptions

E-Mail Address Policies

Exchange Connectors

Exchange Server Certificates

Exchange Servers

Exchange Virtual Directories

Federated Sharing

Information Rights Management

Journaling

Legal Hold

Mail Enabled Public Folders

Mail Recipient Creation

Mail Recipients

Mail Tips

Message Tracking

Migration
Monitoring
Move Mailboxes
Org Custom Apps
Org Marketplace Apps
Organization Client Access
Organization Configuration
Organization Transport Settings
POP3 And IMAP4 Protocols
Public Folders
Receive Connectors
Recipient Policies
Remote and Accepted Domains
Retention Management
Role Management
Security Admin
Security Group Creation and Membership
Security Reader
Send Connectors
Team Mailboxes
Transport Agents
Transport Hygiene
Transport Queues
Transport Rules
UM Mailboxes
UM Prompts
Unified Messaging
User Options
View-Only Audit Logs
View-Only Configuration
View-Only Recipients
WorkloadManagement

- Are you able to add or remove roles from this group?

Yes. If you are logged in as Administrator and the Administrator is a member of the Organization Management role group, you should be able to add or remove management roles from this group.

- What scope (Organizational Units, servers, databases) does this group apply to?

The Organization Management role group applies to the entire Exchange organization. It is not limited to a specific Organizational Unit, server, or mailbox database. Its scope includes all recipients, all Exchange servers, and organization-wide Exchange configuration.

9. Switch to the **User Roles tab** (under **Permissions**).

10. Review the list of existing **Role Assignment Policies**.

11. Answer:

- What is the **default role assignment policy**?

This policy allows you to customize user access when they connect to Outlook on the Web (OWA) options. It defines what settings users are allowed to modify themselves.

- What **roles** are included in the Default Role Assignment Policy?

My Custom Apps

My Marketplace Apps

My ReadWriteMailbox Apps

MyBaseOptions

MyTextMessaging

MyVoiceMail

MyTeamMailboxes

12. Click on the **Default Role Assignment Policy**.

13. Review:

- Which user rights are granted (e.g., change password, manage contacts, configure voicemail settings)?

Users can manage their own mailbox options, contact/profile information, distribution group membership, personal retention settings, mail subscriptions, text messaging, voicemail settings, and some delegation options. It does not grant administrative rights over other users or the Exchange organization.

- Are users allowed to manage distribution groups from Outlook or Outlook Web App (OWA)?

This role enables individual users to view and modify their membership in distribution groups in an organization, provided that those distribution groups allow manipulation of group membership.

Exercise 5 – Exploring Permissions in the Exchange Administration Center (EAC)

Objective:

Understand how auditing is implemented in Exchange Server 2019, including:

- Administrative Audit Logging (actions performed by administrators),
- Mailbox Audit Logging (access to user mailboxes).

Tasks:

Part 1 — Exploring Administrative Audit Logging

1. Open **Exchange Management Shell (EMS)**.
2. Verify if **Administrative Audit Logging** is **enabled** and identify the current configuration settings.

```

ad15 x Ex-Client15
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Get-AdminAuditLogConfig

RunspaceId           : 70b2d610-ac6a-4ca0-a9ff-6d96cacbe00d
AdminAuditLogEnabled  : True
LogLevel             : None
TestCmdletLoggingEnabled : False
AdminAuditLogCmdlets  : {*}
AdminAuditLogParameters : {*}
AdminAuditLogExcludedCmdlets : {}
AdminAuditLogAgeLimit : 90.00:00:00
LoadBalancerCount     : 3
RefreshInterval       : 10
PartitionInfo         : {}
UnifiedAuditLogIngestionEnabled : False
UnifiedAuditLogFirstOptInDate :
AdminDisplayName      :
ExchangeVersion       : 0.10 (14.0.100.0)
Name                  : Admin Audit Log Settings
DistinguishedName     : CN=Admin Audit Log Settings,CN=Global Settings,CN=AD15-Exchange,CN=Microsoft Exchange,CN=Services,CN=Configuration,DC=domain15,DC=local
Identity              : Admin Audit Log Settings
Guid                  : adde3c43-3334-4913-b92c-14ff0903fc14
ObjectCategory        : domain15.local/Configuration/Schema/ms-Exch-Admin-Audit-Log-Config
ObjectClass            : {top, msExchAdminAuditLogConfig}
WhenChanged           : 6/4/2026 12:50:35 PM
WhenCreated           : 6/4/2026 12:30:21 PM
WhenChangedUTC        : 6/4/2026 4:50:35 PM
WhenCreatedUTC        : 6/4/2026 4:30:21 PM
OrganizationId        :
Id                    : Admin Audit Log Settings
OriginatingServer     : ad15.domain15.local
IsValid               : True
ObjectState           : Unchanged

```

- Review the contents of the **Administrative Audit Log** by searching for recent administrative actions.

```

ad15 x Ex-Client15
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Search-AdminAuditLog -StartDate (Get-Date).AddDays(-7) -EndDate (Get-Date) | Format-List

RunspaceId           : 70b2d610-ac6a-4ca0-a9ff-6d96cacbe00d
ObjectModified       : domain15.local/Users/Elon Musk
CmdletName           : Set-Mailbox
CmdletParameters     : {ProhibitSendQuota, ProhibitSendReceiveQuota, Identity, UseDatabaseQuotaDefaults, IssueWarningQuota}
ModifiedProperties    : {}
Caller               : Administrator@domain15.local
ExternalAccess       : False
Succeeded            : True
Error                :
RunDate              : 6/5/2026 11:25:17 AM
OriginatingServer    : AD15 (15.02.1748.010)
Identity             : AAFKAC0KkN2I4ZjhmLlU3YzktNGFhOC05MzFjLTgxMjhhYjYjMwQDUxNGBGAIAAAACoIuy6MmW0QKfCE3Hw0/suBwDmAlM1aOPRSotQ3Fzydva3AAAAATQoAADmAlM1aOPRSotQ3Fzydva3AAAAANEMHAAA=
IsValid              : True
ObjectState          : New

RunspaceId           : 70b2d610-ac6a-4ca0-a9ff-6d96cacbe00d
ObjectModified       : domain15.local/Users/Elon Musk
CmdletName           : New-Mailbox
CmdletParameters     : {LastName, UserPrincipalName, Alias, Password, FirstName, Name}
ModifiedProperties    : {}
Caller               : Administrator@domain15.local
ExternalAccess       : False
Succeeded            : True
Error                :
RunDate              : 6/5/2026 11:22:48 AM
OriginatingServer    : AD15 (15.02.1748.010)
Identity             : AAFKAC0KkN2I4ZjhmLlU3YzktNGFhOC05MzFjLTgxMjhhYjYjMwQDUxNGBGAIAAAACoIuy6MmW0QKfCE3Hw0/suBwDmAlM1aOPRSotQ3Fzydva3AAAAATQoAADmAlM1aOPRSotQ3Fzydva3AAAAANEMHAAA=

```

- Perform a basic administrative action (e.g., modify a mailbox setting).

```
ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Set-Mailbox "Elon Musk" -CustomAttribute1 "AuditTest"
```

5. Search the **Administrative Audit Log** again and verify that your action has been recorded.

```
ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Search-AdminAuditLog -StartDate (Get-Date).AddHours(-1) -EndDate (Get-Date) -Cmdlets Set-Mailbox

RunspaceId      : ee37dadb-b2ec-4849-8225-4579fa548aef
ObjectModified   : domain15.local/Users/Elon Musk
CmdletName       : Set-Mailbox
CmdletParameters : {Identity, CustomAttribute1}
ModifiedProperties : {}
Caller           : Administrator@domain15.local
ExternalAccess    : False
Succeeded         : True
Error            :
RunDate          : 6/5/2026 12:50:35 PM
OriginatingServer : AD15 (15.02.1748.010)
Identity         : AAMKADNKN2I4ZjhMLTU3YzktNGFhOC05MzFjLTgxMjhhYjMwODUxNgBGAAAAAcoIuy6Wrw0QKFcE3HwO/sUBWDMAlw1aOPRSotQJ
Invalid          : True
ObjectState      : New
```

Part 2 — Exploring Mailbox Audit Logging

6. Check if **Mailbox Audit Logging** is enabled for a specific user mailbox.

```
ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Get-Mailbox "Elon Musk" | Format-List AuditEnabled

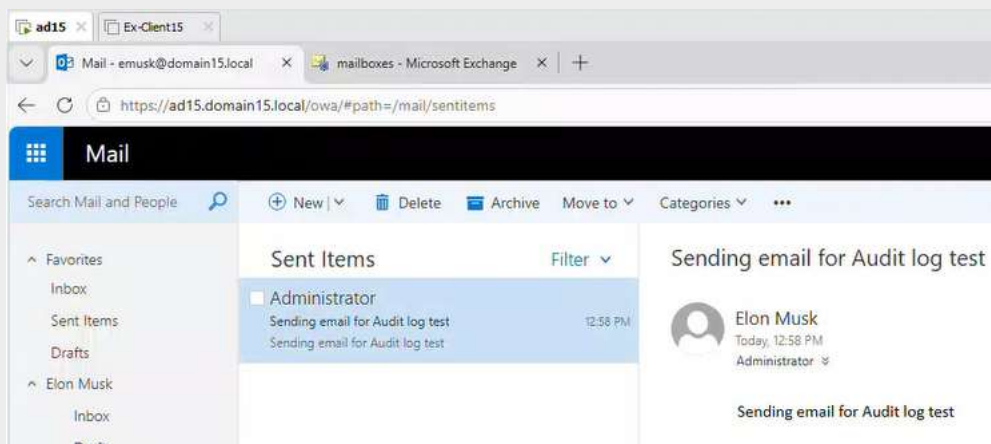
AuditEnabled : False
```

7. If auditing is not enabled, enable **Mailbox Audit Logging** for the user.

```
ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Set-Mailbox "Elon Musk" -AuditEnabled $true
[PS] C:\Users\Administrator>Get-Mailbox "Elon Musk" | Format-List AuditEnabled

AuditEnabled : True
```

8. Simulate mailbox activity by logging into the user's mailbox and performing an action (e.g., sending or reading an email).



9. Search the **Mailbox Audit Log** to find and review the mailbox activities performed.

```

[PS] C:\Users\Administrator>New-MailboxAuditLogSearch -Mailboxes "emusk@domain15.local" -LogonTypes Owner,Delegate -ShowDe
Date).AddDays(-60) -EndDate (Get-Date) -StatusMailRecipients "administrator@domain15.local"

SerializationData      : {35, 115, 105, 103, 35, 45, 50, 0, 0, 0, 1, 0, 0, 0, 255, 255...}
RunspaceId             : ee37dadb-b2ec-4849-8225-4579fa548aef
Mailboxes              : {domain15.local/Users/Elon Musk}
LogonTypes             : {Owner, Delegate}
Operations             : {}
ShowDetails           : True
HasAttachments         : 
ConsumerMailbox        : False
Name                  : Search20260605{6e1303dd-b8be-458d-bcff-0d677dceae9a}
StartDateUtc          : 4/6/2026 5:09:47 PM
EndDateUtc            : 6/5/2026 5:09:47 PM
StatusMailRecipients  : {administrator@domain15.local}
CreatedBy             : domain15.local/Users/Administrator
ExternalAccess        : 
QueryComplexity        : 0
Identity              : 0e510acf-3881-4ecb-b872-36790c3ff0e7
IsValid               : True
ObjectState           : New
  
```